



**YEAR 11 Mathematics
Preliminary Course
Assessment Task 2
June 2011
Paper 1**

1. There are 3 sections.
2. Answer each question on your own paper showing all necessary working
3. Start each section on a new page
4. Calculators may be used

Topic	Mark
1. Section 1 (Quadratic Polynomial)	/18
2. Section 2 (Quadratic Polynomial)	/14
3. Section 3 (Geometry- angles, polygons)	/26

TOTAL /58

Section 1 Paper 1 (Quadratic Polynomial)

- 1) Solve the equation $2x^2 - 5x + 1 = 0$ leaving your answer in surd form.
 - 2) Determine if the roots of the quadratic equation $9x^2 - 6x + 1 = 0$ are real or unreal, rational or irrational, equal or unequal.
 - 3) Without solving, show that the roots of the quadratic equation $2x^2 - 4x + 9 = 0$ are unreal.
 - 4) For what values of r does the equation $rx^2 - (3r + 1)x + r = 0$ have equal roots?
 - 5) Without sketching, show that the quadratic function $f(x) = 2x^2 - 3x + 7$ lies entirely above the x -axis.
 - 6) For what values of r will the quadratic expression $rx^2 - 4x - 7$ be negative definite?
 - 7) Find a quadratic function $f(x) = ax^2 + bx + c$, which is positive for $\sqrt{2} < x < 3\sqrt{2}$ only.
 - 8) Find a quadratic equation for which the sum of the roots is -10 and their product is 24 .
 - 9) Find the value of r for which the roots of the quadratic equation $3x^2 - 4x + r = 0$ are reciprocals of one another.
 - 10) If $x = 3$ is one root of the equation $x^2 + 5x - r^2 = 0$, find the other root and the value of r .
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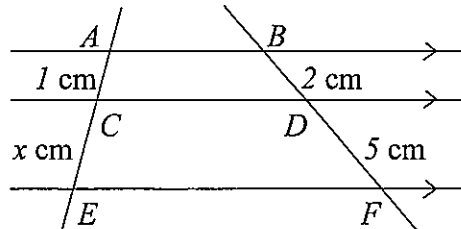
Section 2 Paper 1 (Quadratic Polynomial)

- 1) Let α and β be the roots of the equation $x^2 + 5x - 2 = 0$. Without solving, find the value of:
a. $\alpha + \beta$; b. $\alpha\beta$; c. $\alpha^2\beta + \alpha\beta^2$; d. $\alpha^3 + \beta^3$
 - 2) Solve $x^4 - x^2 - 2 = 0$
 - 3) Solve $9^x - 4(3^x) + 3 = 0$
 - 4) Rewrite the expression $x^2 - 4x - 3$ in the form $a(x - 1)^2 + b(x + 2)$
 - 5) Find the values of m for which $12 + 4m - m^2 > 0$
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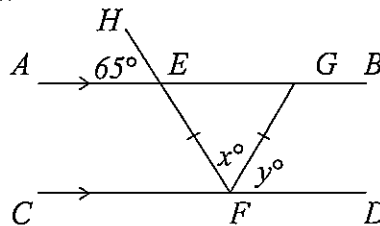
Section 3 Paper 1 (Geometry – Angles and polygons)

Reasons **NOT** required for questions 1 to 9

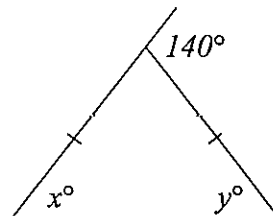
- 1) Use the information in the diagram given below to find the value of x .



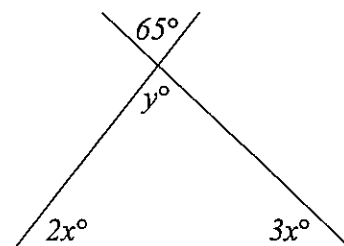
- 2) In the diagram given below, $AB \parallel CD$ and $EF = GF$. Find the values of x and y if $\angle AEH = 65^\circ$.



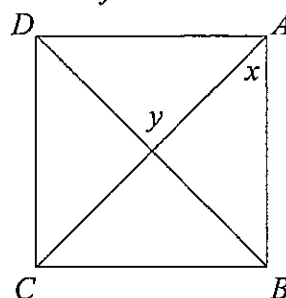
- 3) Use the information in the figure given below to find the values of x and y .



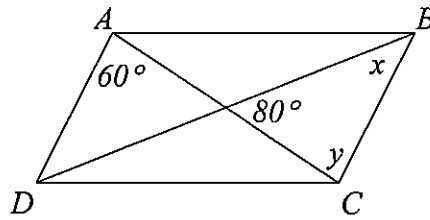
- 4) Use the information in the figure given below to find the values of x and y .



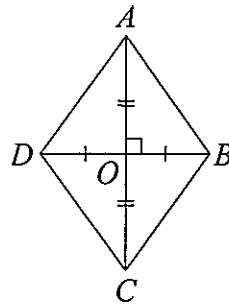
- 5) $ABCD$ is a square. Find the values of x and y .



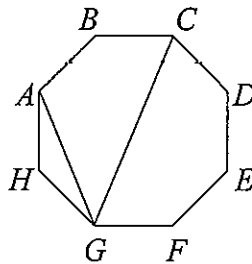
- 6) $ABCD$ is a parallelogram. Find the values of x and y .



- 7) Find the area and perimeter of the rhombus $ABCD$, given that $AO = 8$ cm and $BO = 6$ cm.



8)

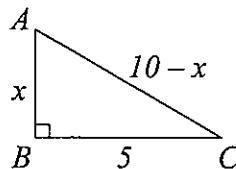


NOT TO
SCALE

$ABCDEFGH$ is regular octagon.

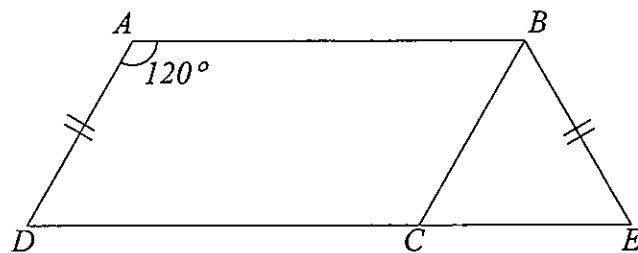
- Calculate the size of $\angle ABC$.
- Calculate the size of $\angle GAH$.
- Using (i), or otherwise, calculate the size of $\angle CGF$.
- Hence, calculate the size of $\angle AGC$.

9)



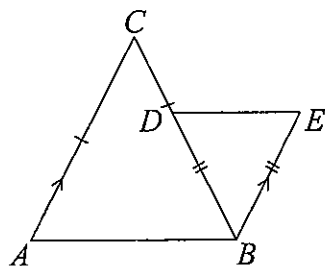
In the diagram $\angle ABC$ is a right angle. Find the value of x .

10)



The diagram shows a parallelogram $ABCD$ with $\angle DAB = 120^\circ$. The side DC is produced to E so that $AD = BE$. Prove that $\triangle BCE$ is equilateral.

11)



In the diagram above, $AC = BC$, $BD = BE$ and $AC \parallel BE$. Prove that $AB \parallel DE$.



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Topic	Mark
1. Section 1 (Linear Functions)	/18
2. Section 2 (Linear Functions)	/14
3. Section 3 (Geometry- Congruency, similarity)	/20

TOTAL /52

Section 1 Linear Functions

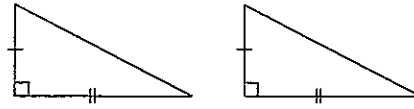
- 1) Express the equation of the straight line $y = 2 - \frac{x}{3}$ in general form.
 - 2) Write down the gradient and y -intercept of the line $5x - 3y + 2 = 0$.
 - 3) Show that the point $(-3, -7)$ lies on the line $5x - 2y + 1 = 0$.
 - 4) Sketch the graph of $y = 2x + 1$, indicating clearly where the line cuts the x and y axes.
 - 5) Sketch the graph of $y = 3$.
 - 6) Find the gradient of the straight line passing through the points $(-5, 6)$ and $(-2, 3)$.
 - 7) Find (to 1 decimal place) the gradient of the straight line with an inclination of 97° with the positive direction of the x axis.
 - 8) Find the length of the interval joining $(4, 3)$ and $(7, 7)$.
 - 9) Find the co-ordinates of the mid-point of the interval joining $(2, 1)$ and $(4, 7)$.
 - 10) The mid-point of the interval joining $(2x + 1, 1 + y)$ and $(4y + 1, 3 + y)$ is $(5, 1)$. Find x and y .
 - 11) Find the equation of the straight line (in gradient-intercept form) which passes through $(2, 3)$ and has a gradient of 1 .
 - 12) Find the equation of the straight line (in general form) which passes through the points $(3, 2)$ and $(5, 3)$.
-

Section 2 Linear Functions

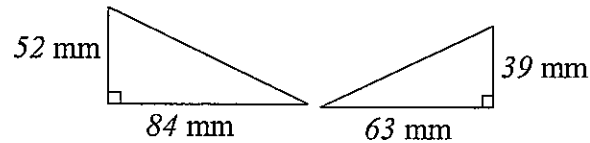
- 1) Find the equation of the straight line passing through $(-3, 1)$ and parallel to $6x - 12y + 5 = 0$.
- 2) Find the equation of the straight line passing through $(5, 2)$ and the point of intersection of the lines $3x - y - 5 = 0$ and $3x + y - 7 = 0$.
- 3) Find the perpendicular distance of the point $(5, 4)$ from the straight line $3x + 4y + 9 = 0$.
- 4) Show (without sketching on the number plane) that the points $(-2, 2)$ and $(3, -2)$ lie on opposite sides of the straight line $2x - y + 2 = 0$.
- 5) $P(1, 2)$ is a point on the line $3x + y - 5 = 0$. Find the coordinates of the points on this line so that their distance from P is $\sqrt{10}$ units.

Section 3 Geometry (Congruence and similarity)

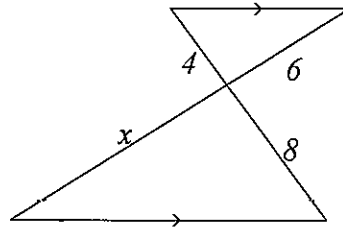
- 1) The triangles below are congruent. State the condition for the congruency (*SSS*, *SAS*, *AAS* or *RHS*).



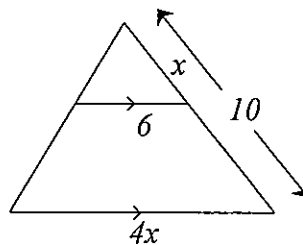
- 2) Are the triangles below similar? Give a reason for your answer.



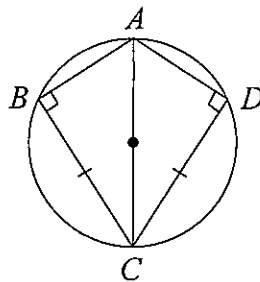
- 3) In the diagram given below, find the value of x . (All lengths are in cm.)



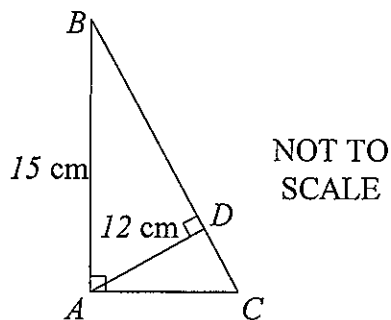
- 4) In the diagram given below, find the value of x . (All lengths are in cm.)



- 5) Prove that $\triangle ABC \equiv \triangle ADC$.

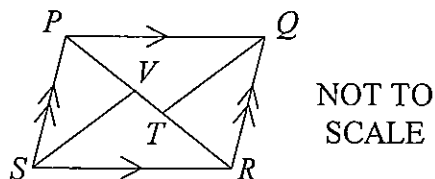


- 6) $\triangle ABC$ is right-angled at A and AD is drawn perpendicular to BC . $AB = 15$ cm and $AD = 12$ cm. Copy the given diagram onto your answer sheet.



- i. Show that $BD = 9$ cm.
- ii. Prove that $\triangle ABC$ is similar to $\triangle DBA$.
- iii. Hence find the length of AC .

7)



$PQRS$ is a parallelogram. TQ bisects $\angle PQR$ and VS bisects $\angle PSR$.

- i. Copy this diagram onto your answer booklet.
- ii. State why $\angle PQR = \angle PSR$.
- iii. Prove that $\triangle PVS$ and $\triangle RTQ$ are congruent.
- iv. Hence find the length of TV if $PR = 20$ cm and $TR = 8$ cm

Year 11 (Paper 1)
Mathematics (Solutions)

$$\begin{aligned} \text{Q1/ } x &= \frac{5 \pm \sqrt{25 - 4 \times 1 \times 2}}{4} \\ &= \frac{5 \pm \sqrt{17}}{4} \end{aligned} \quad (1)$$

$$\begin{aligned} \text{Q2/ } \Delta &= 36 - 4 \times 9 \times 1 \\ &= 0 \end{aligned}$$

$\therefore$ Roots are real, rational & equal (1)

$$\begin{aligned} \text{Q3/ } \Delta &= 16 - 4 \times 9 \times 2 \\ &= - \end{aligned}$$

$\therefore \Delta < 0$ Roots are unreal (1)

$$\begin{aligned} \text{Q4/ when } \Delta &= 0: \\ (3r+1)^2 - 4r^2 &= 0 \\ 9r^2 + 6r + 1 - 4r^2 &= 0 \\ 5r^2 + 6r + 1 &= 0 \\ \begin{array}{cc} 5r & 1 \\ r & 1 \end{array} & \quad \text{Crossed out} \\ (5r+1)(r+1) &= 0 \\ r = -\frac{1}{5} \text{ or } -1 & \end{aligned} \quad (2)$$

25) $a = 2$ $\therefore a > 0$
 $b^2 - 4ac = 9 - 4 \times 2 \times 7$

$$=$$

(2)

$$\therefore \Delta < 0$$

$\therefore$ positive definite

$\therefore$ lies entirely above x axis

26/ $r < 0$ and $\Delta < 0$

$$\Delta = 16 - 4 \times r \times -7$$

$$\Delta = 16 + 28r$$

$$16 + 28r < 0$$

$$28r < -16$$

$$r < \frac{-16}{28}$$

$$28$$

(2)

$$r < -\frac{4}{7}$$

$\therefore r < -\frac{4}{7}$ to be neg def.

Q7/ $x^2 - (\sqrt{2} + 3\sqrt{2})x + \sqrt{2} \cdot 3\sqrt{2} = 0$

$$x^2 - 4\sqrt{2}x + 6 = 0$$

But can come down

(2)

so

$$-x^2 + 4\sqrt{2}x - 6 = 0$$

Q8/ $x^2 + 10x + 24 = 0$

(2)

Q9/ $2 \times \frac{1}{2} = \frac{r}{3}$

$$1 = \frac{r}{3}$$

$$r = 3$$

(2)

$$10/ \quad 9 + 15 - r^2 = 0$$

$$r^2 = 24$$

$$r = \pm 2\sqrt{6}$$

$$3 + \beta = -5$$

(3)

$$\beta = -8$$

$\therefore$ other root is -8

Section 2 Paper 1

$$1/ (a) -5$$

(1)

$$(b) -2$$

(1)

$$(c) \alpha\beta(\alpha+\beta)$$

$$-2 \times -5$$

$$= 10$$

(2)

$$(d) (\alpha+\beta)(\alpha^2 - \alpha\beta + \beta^2)$$

$$(\alpha+\beta)[(\alpha+\beta)^2 - 2\alpha\beta - 2\alpha\beta]$$

$$(\alpha+\beta)[(\alpha+\beta)^2 - 3\alpha\beta]$$

$$(-5)[25 - 3 \times -2]$$

(2)

$$= -155$$

$$2/ \quad x^4 - x^2 - 2 = 0$$

$$\begin{array}{ccc} x^2 & & -2 \\ & \times & \\ x^2 & & +1 \end{array}$$

$$(x^2 - 2)(x^2 + 1) = 0$$

(2)

$$x = \pm \sqrt{2}$$

$\Rightarrow$ no real

solutions

$$\text{for } x^2 + 1 = 0$$

4/ Let $u = 3^x$

$$u^2 - 4u + 3 = 0$$

$$(u-3)(u-1) = 0$$

$$u = 3, 1$$

$$3^x = 3 \quad \text{and} \quad 3^x = 1$$

$$x = 1$$

$$x = 0$$

5/ $x^2 - 4x - 3 \equiv a(x^2 - 2x + 1) + bx + 2b$

$$\equiv ax^2 - 2ax + a + bx + 2b$$

$$\therefore a = 1$$

$$\equiv x^2 - 2x + 1 + bx + 2b$$

$$\therefore 2b + 1 = -3$$

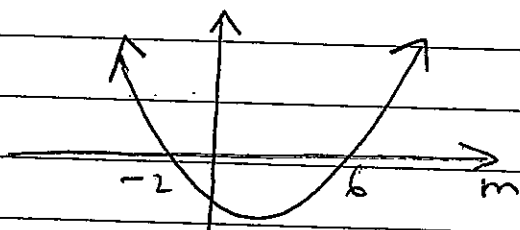
$$2b = -4$$

$$b = -2$$

$$\therefore x^2 - 4x - 3 \equiv (x-1)^2 - 2(x+2)$$

6/ $m^2 - 4m - 12 < 0$

$$(m-6)(m+2) < 0$$



$$\therefore -2 < m < 6$$

Geometry

$$1/ \quad \frac{x}{1} = \frac{5}{2} \quad (2)$$
$$x = 2\frac{1}{2}$$

$$2/ \quad x = 50^\circ, y = 65^\circ \quad (2)$$

$$3/ \quad x = y = 70^\circ \quad (2)$$

$$4/ \quad y = 65^\circ$$
$$5x = 115$$
$$x = 23^\circ \quad (2)$$

$$5/ \quad x = 45^\circ$$
$$y = 90^\circ \quad (2)$$

$$6/ \quad y = 60^\circ$$
$$x = 40^\circ \quad (2)$$

$$7/ \quad \text{Perimeter} = 40 \text{ cm}$$
$$\text{Area} = 96 \text{ cm}^2 \quad (2)$$

$$8/ (i) \quad \frac{6 \times 180}{8} = 135^\circ$$
$$\therefore \angle ABC = 135^\circ \quad (1)$$

$$(ii) \quad \frac{180 - 135}{2} = 22\frac{1}{2}^\circ \quad (1)$$

$$(iii) \quad 67\frac{1}{2}^\circ \quad (1)$$

$$(iv) \quad 135 - 22\frac{1}{2} - 67\frac{1}{2} = 45 \quad (1)$$

$$9/ \quad (10 - x)^2 = x^2 + 25$$
$$100 - 20x + x^2 = x^2 + 25$$
$$75 = 20x \quad x = 3.75 \quad (2)$$

10/ $\angle DCB = 120^\circ$ (opp $\angle$'s parallelogram =)

$\angle BCE = 60^\circ$ (adj supp $\angle$)

$AD = BC$ (opp sides parallelogram =)

$\therefore BC = BE$ (given $AD = BE$)

$\therefore \triangle BCE$ is isosceles

as $\angle BCE = \angle BEC = 60^\circ$

$\therefore \triangle BCE$ is equilateral

11/ let $\angle ACB = x$

$\therefore \angle CBE = x$ (alt $\angle$'s $AC \parallel BE$)

$\angle BDE = \frac{180 - x}{2}$ (base $\angle$'s = isoc. $\triangle BDE$)

$\angle CBA = \frac{180 - x}{2}$ (base $\angle$'s = isoc. $\triangle ACB$)

$\therefore \angle BDE = \angle CBA$ and are in alt. positions

$\therefore AB \parallel DE$

Paper two

Section 1 Linear Functions

$$1) \quad 3y = 6 - x$$

$$x + 3y - 6 = 0 \quad (1)$$

$$2) \quad 3y = 5x + 2$$

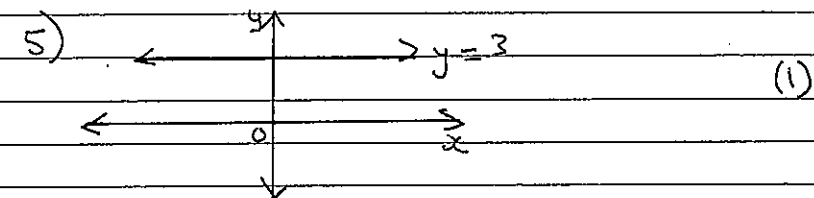
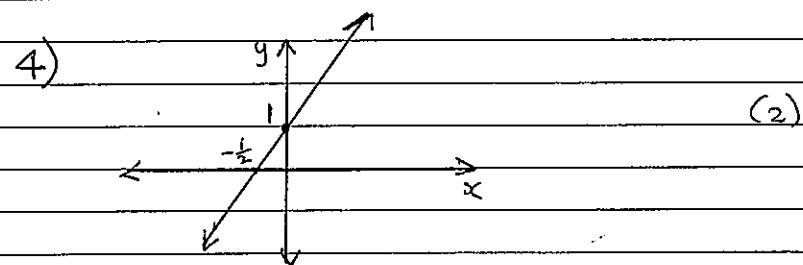
$$y = \frac{5x + 2}{3} \quad \text{gradient} = \frac{5}{3}$$

$$y \text{ intercept} = \frac{2}{3} \quad (2)$$

$$3) \quad 5(-3) - 2(-7) + 1 = 0$$

$$-15 + 14 + 1 = 0$$

$$0 = 0 \quad (1)$$



$$6) \quad m = \frac{6-3}{-5+2}$$

$$= \frac{3}{-3}$$

$$m = -1 \quad (1)$$

$$7) \quad m = \tan 97^\circ$$

$$m = -8.1 \quad (2)$$

$$8) \quad d = \sqrt{(4-7)^2 + (3-7)^2}$$

$$= \sqrt{(-3)^2 + (-4)^2} \quad (1)$$

$$= \sqrt{9+16}$$

$$= 5$$

$$9) \quad \left(\frac{2+4}{2}, \frac{1+7}{2} \right) = (3, 4) \quad (1)$$

$$10) \quad \frac{2x+1+4y+1}{2} = 5$$

$$2x+4y+2 = 10$$

$$2x+4y = 8$$

$$x+2y = 4 \quad \text{--- (1)}$$

$$\frac{1+y+3+y}{2} = 1 \quad (2)$$

$$2y+4 = 2$$

$$2y = -2$$

$$y = -1$$

Sub $y = -1$ into (1)

$$x - 2 = 4$$

$$x = 6$$

$$11) \quad y - 3 = 1(x - 2) \quad (2)$$

$$y - 3 = x - 2$$

$$y = x + 1$$

$$12) \quad m = \frac{-1}{-2} \quad y - 2 = \frac{1}{2}(x - 3)$$

$$m = \frac{1}{2} \quad 2y - 4 = x - 3 \quad (2)$$

$$0 = x - 2y + 1$$

13) $y = -5x + 2$
 i. line parallel has gradient of -5 (1)

14) $m_{\perp} = 3$

Section 2 Linear Functions

1) $12y = 6x + 5$
 $y = \frac{1}{2}x + \frac{5}{12}$

$y - 1 = \frac{1}{2}(x + 3)$

$2y - 2 = x + 3$

$0 = x - 2y + 5$

OR

$y = \frac{x}{2} + \frac{5}{2}$

(2)

2) $3x + y = 7$
 $3x - y = 5$ } Add

$6x = 12$

$x = 2$

$6 - y = 5$

$y = 1$

$m = \frac{1-2}{2-5}$

$= \frac{-1}{-3}$

$= \frac{1}{3}$

$y - 1 = \frac{1}{3}(x - 2)$

(3)

$3y - 3 = x - 2$

$0 = x - 3y + 1$

OR

$y = \frac{x}{3} + \frac{1}{3}$

3) $d = \frac{|(3)(5) + (4)(4) + 9|}{\sqrt{9+16}}$

$= \frac{|15 + 16 + 9|}{5}$

$= \frac{40}{5}$

(2)

$= 8 \text{ units}$

4) For $(-2, 2)$

$d_1 = (2)(-2) + (-1)(2) + 2$
 (numerator)

$= -4 - 2 + 2$

$= -4$

For $(3, -2)$

$d_2 = (2)(3) + (-1)(-2) + 2$

(numerator) $= 6 + 2 + 2$

$= 10$

(3)

As the numerator (without absolute value) of the distance formula gives values opposite in sign, the points are on opposite sides

5) $\sqrt{(x-1)^2 + (y-2)^2} = \sqrt{10}$

$(x-1)^2 + (y-2)^2 = 10$

$x^2 - 2x + 1 + y^2 - 4y + 4 = 10$

$x^2 - 2x + y^2 - 4y = 5$ --- (1)

For $3x + y - 5 = 0$

$y = -3x + 5$

$y = 5 - 3x$ --- (2)

Sub (2) into (1)

$$x^2 - 2x + (5-3x)^2 - 4(5-3x) = 5$$

$$x^2 - 2x + 25 - 30x + 9x^2 - 20 + 12x = 5$$

$$10x^2 - 20x = 0$$

$$10x(x-2) = 0$$

$$x = 0, 2$$

$$y = 5, -1$$

(4)

∴ Points are (0, 5) and (2, -1)

Section 3 Geometry

1) SAS

(1)

$$2) \frac{52}{39} = 1\frac{1}{3} \quad \frac{84}{63} = 1\frac{1}{3}$$

∴ Δ's are similar as two sides
in ratio, included ∠ equal

(2)

$$3) \frac{4}{8} = \frac{6}{x}$$

$$4x = 48$$

$$x = 12$$

(1)

$$4) \frac{x}{10} = \frac{6}{4x}$$

$$4x^2 = 60$$

$$x^2 = 15$$

$$x = \sqrt{15}$$

(2)

$$5) BC = DC \text{ (given)}$$

$$\angle ABC = \angle ADC \text{ (given)}$$

AC is common

$$\therefore \triangle ABC \equiv \triangle ADC \text{ (RHS)} \quad (2)$$

$$6) (i) (BD)^2 = 15^2 - 12^2$$

$$(BD)^2 = 81$$

$$BD = 9$$

(1)

$$(ii) \angle BAC = \angle BDA = 90 \text{ (given)}$$

$$\angle ABC = \angle ABD \text{ (common } \angle)$$

$$\therefore \triangle ABC \equiv \triangle DBA \text{ (Equiangular)} \quad (3)$$

$$(iii) \frac{15}{9} = \frac{AC}{12}$$

$$9AC = 180$$

$$AC = 20 \text{ cm}$$

(2)

$$7) (ii) \text{ opp } \angle \text{ s parallelogram are equal} \quad (1)$$

$$(iii) \angle PSV = \angle TQR \text{ (bisected)}$$

Equal angles of parallelogram

$$\angle SPV = \angle QRT \text{ (alt } \angle \text{ s } PS \parallel QR)$$

$$PS = QR \text{ (opp sides of parallelogram are equal)}$$

(4)

$$\therefore \triangle PVS \equiv \triangle RTQ \text{ (AAS)}$$

$$(iv) 20 - 16 = 4 \text{ cm}$$

(1)